

NTT Basic Research Laboratories (BRL) Socio-economic report

Industrial R&D Internship

NTT Basic Research Laboratories, NTT Inc.
Atsugi R&D Center, Japan
Advanced Applied Physical Science Laboratory
Nanodevices Research Group

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July 2025 - December 2025

Introduction

This report analyzes the organization, financing, management practices and social environment of the research group at NTT BRL where I completed my internship. This report was prepared using direct observation and participation during my internship at NTT BRL, informal interviews and discussions with group members and my supervisor, internal documents shared during the internship, and public NTT disclosures and BRL web pages. The aim is to combine descriptive reporting with critical analysis tied to evidence and firsthand experience.

Some key findings are reported in this document. (i) BRL combines high technical autonomy with active mentorship but faces the common trade-off between exploratory research and short-term deliverables, (ii) R&D financing is largely internal with the corporate budget and (iii) sustainability and occupational safety practices exist but could better quantify R&D-specific environmental footprints. The report ends with recommendations for improved documentation, sustainability actions, and a personal professional development plan.

The report is structured as follows. Section 1 gives a concise overview of the NTT Group and the role of the Basic Research Laboratories. Section 2 describes the company's working environment, including typical duties of research engineers, management and working habits, diversity and career perspectives, and occupational health and safety. Section 3 examines sustainability and environmental considerations, including NTT's targets, R&D programs and participation in Expo 2025. Finally, Section 4 lays out my personal review.

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Chapter 1

About NTT Group

1.1 Worldwide influence

NTT Group is a leading global company in telecommunications and information-and-communications technology (ICT). Headquartered in Tokyo, Japan, the Group combines an extensive domestic footprint with a broad global presence: as of March 31, 2025 NTT employs roughly 340,000 people, operates through over 900 consolidated subsidiaries and maintains a local presence in more than 80 countries and regions [1]. Ranked 55th in the Fortune Global 500, NTT is the fourth largest telecommunications company in the world in terms of revenue [2], and was the third largest publicly traded company in Japan after Toyota and Sony as of June 2022 [3].

The Group plays a central role in Japan’s digital and communications infrastructure. Its global activities span over 80 countries, supported by a comprehensive RD apparatus focused on pushing the boundaries of information and communication technologies (ICT). With annual revenues around USD 90 billion and a long-term technology vision known as IOWN (Innovative Optical and Wireless Network), NTT positions itself as an innovation-driven actor addressing societal, industrial, and environmental challenges of the coming decades.

1.2 Company’s roots

The origins of NTT trace back to the postwar period of Japanese reconstruction. The Nippon Telegraph and Telephone Public Corporation Act established a national public telecommunications entity in 1952, based on an earlier telegraph connection between Tokyo and Yokohama that dated to 1870. In 1985 the public corporation was privatized to open Japan’s telecom market to competition and accelerate technological and business innovation, making Japan the second country in the world, after the United States, to deregulate its telecom market. Following privatization, the company reorganized by creating specialized subsidiaries for different business lines and by establishing the NTT Group structure that gradually evolved into today’s collection of operating companies and service brands [1].

The same year of 1985 saw the creation of the Basic Research Laboratories (BRL), one of NTT’s cornerstone research organizations. BRL was established to conduct long-term, fundamental scientific research that would seed future technological breakthroughs. Over the past decades, its contributions in materials science, quantum information, semiconductor devices, photonics, and condensed-matter physics have supported several of NTT’s strategic innovation domains.

In 2024–2025, NTT undertook a Corporate Identity (CI) renewal, updating branding, strategic messaging, and organizational communication frameworks to reflect the digital, global, and sustainability-oriented direction of the Group.

1.3 NTT Group’s subsidiaries

NTT Group is organized so that its major operating businesses can focus on specific markets and services while sharing common technology and capital resources.

Subsidiary	Purpose
NTT East / NTT West.	Regional carriers that provide fixed-line, broadband and business communications services across Japan's eastern and western regions.
NTT DOCOMO Group.	The mobile and consumer-facing arm, which also runs enterprise ICT initiatives and is central to the Group's 5G and DX (digital transformation) efforts.
NTT DATA Group.	The Group's global IT services and consulting business: application development and management, system integration, data-center services, managed services and industry-specific solutions.
NTT Urban Solutions / NTT Anode Energy.	Businesses that combine urban development, energy solutions (including renewables and energy storage) and next-generation computing/space-computing services.
NTT R&D.	The research and innovation pillar, composed of several laboratories, including the Basic Research Laboratories (BRL), the Device Innovation Center, the Network Service Systems Labs, and the IOWN Integrated Innovation Center.

Table 1.1 – Several key business segments responsible for the NTT Group's value chain.

1.3.1 About NTT Research & Development (R&D)

NTT makes substantial investments in research: the Group dedicates roughly USD 3 billion, which is approximately 3.3% of total annual revenue, and employs about 2,300 research professionals. NTT's R&D portfolio spans materials science, photonics, quantum information, networking architectures and software platforms, and it explicitly targets social challenges such as security, disaster resilience and sustainability while pursuing new infrastructure concepts like IOWN (Innovative Optical and Wireless Network) [4, 5]. These investments and the Group's patent and publication output, which is approximately 22,000 patents to date, including some 14,000 registered overseas, underscore NTT's long-term, science-led approach to innovation [7]. It is important to note that very few other telecommunications companies maintain comparably broad, in-house basic research divisions.

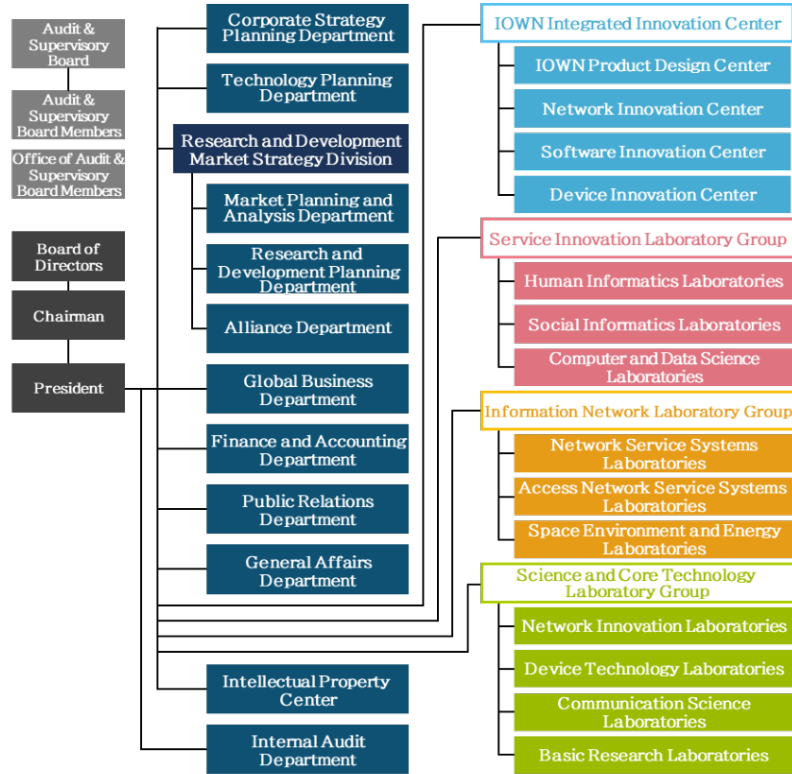


FIG-2

Figure 1.1 – Organization Chart of Nippon Telegraph and Telephone Corporation [6].

A distinctive characteristic of NTT's R&D philosophy is its dual-track approach:

- Long-term basic research (10–20-year horizon) conducted mainly at BRL and other foundational laboratories.
- Applied and translational research aligned with business units, often leading to direct commercialization or integration into network infrastructure.

This structure ensures that fundamental scientific discoveries can be matured into technologies supporting the Group's social and industrial missions.

1.3.2 About NTT Basic Research Laboratories (NTT BRL)

NTT Basic Research Laboratories (BRL) belongs to the Group's R&D organization (the Science and Core Technology Laboratory Group) and focuses on long-horizon basic science that ultimately feeds into NTT's technical road map and business lines. It was founded in 1985 at the time of the Group's privatization.



Figure 1.2 – Organization chart of NTT-BRL [6]. I work in the Nanodevices Research Group, part of the Advanced Applied Physical Science Laboratory.

NTT-BRL is divided into different laboratories, each containing several research groups. Groups operate with significant autonomy over experimental plans while aligning with higher-level BRL strategy, aiming to generate scientific *seeds* that may later inspire transformational technologies. BRL is globally recognized for contributions in:

- quantum transport and nanofabricated devices,
- semiconductor heterostructures and strongly correlated materials,
- single-photon sources and quantum optics,

- spintronics and emerging memory technologies,
- low-temperature experimental physics and advanced material characterization.

BRL is evaluated periodically by an external Advisory Board that reviews research plans, assesses achievements, and strengthens cooperative ties between BRL and partner organizations. This external review helps ensure BRL’s research remains rigorous, internationally visible and aligned with both scientific frontiers and long-term industrial opportunities [6].

BRL conducts basic research across Materials Science, Physical Science and Optical Science, while maintaining active collaborations with universities and research institutes worldwide. BRL also organizes international events and educational initiatives such as the *BRL School*, which bring early-career researchers and students into close contact with BRL’s research teams and visiting authorities [6, 7].

The Nanodevices Research Group. Within BRL, the Nanodevices Research Group focuses on the design, fabrication, and characterization of advanced nano-electronic and quantum devices. Research topics include charge transport in low-dimensional systems, single-electron and single-hole devices, superconducting hybrids, quantum dots, and next-generation transistor architectures. The Group combines cleanroom fabrication, cryogenic measurements (down to millikelvin temperatures), custom electronics, and data-driven modeling. Its role fits strategically within NTT’s long-term vision: advancing quantum and nanoelectronic technologies that could underpin future low-power computing architectures, secure communication systems, and components of the IOWN initiative.

1.4 NTT Group’s Corporate Identity

In May 2025 NTT announced a renewal of its corporate identity. From July 1, 2025 the corporate logotype and some company names in the Group were updated globally and the parent company adopted the abbreviated corporate name *NTT, Inc.* (together with a refreshed logotype) as part of a broader move to harmonize branding for international markets. This CI renewal aims to strengthen NTT’s global recognition and to present a unified visual identity as the Group accelerates overseas expansion [8].

Alongside the CI renewal, the Group restated a concise purpose and set of values to guide its strategy and operations. The stated core is: *Innovating a Sustainable Future for People and Planet.* The Group’s values emphasize *integrity, building trust, and creating communications*, principles intended to steer decision-making across a large and diversified global organization. This framing links the Group’s sustainability commitments, its long-term investment in foundational R&D and its business activities [4, 5].



Figure 1.3 – Renewal of NTT Group’s Corporate Identity and NTT Group’s subsidiaries.

Chapter 2

Company's working environment

2.1 Engineer's duty

Although the members of my research group hold scientific academic backgrounds rather than a formal *engineer* track, their day-to-day responsibilities require substantial engineering skills. Researchers in the group perform long-horizon, foundational R&D. They formulate and validate new concepts, design and build experimental prototypes, operate sophisticated measurement setups, run simulations, analyze experimental data, publish results in the academic literature, and contribute ideas that can eventually be transferred into NTT products and services [6, 7]. Both theoretical understanding and practical laboratory craftsmanship are needed to conceive and implement many of the group's self-built experimental systems.

Positions within the group span early-career researcher/engineer roles through to senior and distinguished researchers who lead programs and set technical direction. Performance and career progression are typically evaluated on research output (peer-reviewed papers and patents), technical leadership, and demonstrable impact on long-term business and R&D road maps [1, 6].

In short, researchers, especially those working in my group, must be versatile in order to be able to design and assemble devices, understand the physics that govern them, and process data using numerical and programming tools. These requirements are why I believe that programs such as the one from ESPCI Paris-PSL prepare graduates well for research roles in large-scale technology organizations.

2.2 Working habits

2.2.1 Working management

My supervisor Dr. Gento Yamahata manages me in a very flexible and supportive way. He regularly interrupts his own work to provide advice, answers questions promptly, and gives autonomy for experimental initiative. Typical supervisory practices I experienced include:

- Daily or frequent technical discussions that compare current measurements with prior results and theoretical expectations.
- Regular check-ins on short- and long-term objectives to keep the project focused while preserving experimental freedom.
- Close, hands-on mentoring for complex equipment, such as the dilution refrigerator.
- Freedom to propose and pursue side studies, such as numerical simulations to reproduce device characteristics.
- Discretion to allocate time for writing reports and summarizing results.

Group members beyond the supervisor frequently ask me about my ongoing work, which develops my communication skills and triggers my productive cross-pollination of ideas. Informal office discussions between members are common and, despite my limited Japanese, I felt them as respectful and scientifically useful for both sides.

The group leader, Dr. Katsuhiko Nishiguchi, adopts a light-touch management style. Members regularly consult him for guidance and direction and the group leader does not micromanage day-to-day work at all. He typically

adopts a strategic, high-level oversight role, ensuring that research directions align with the laboratory’s long-term objectives but allowing considerable autonomy to individual researchers.

Larger-scale meetings within the group, across laboratories, or between related research teams are organized regularly to align priorities and stimulate interdisciplinary collaboration [6].

2.2.2 Working conditions

Long working hours and strong hierarchical expectations have long been noted features of work culture in Japan. Excessive overtime can have serious health consequences, a phenomenon known as *karōshi* (death from overwork). Many workers report routinely working beyond the statutory forty-hour week. Commuter trains full of sleeping passengers are a visible symptom of these pressures, especially during periods of high workload. Despite government and corporate efforts to improve work–life balance, cultural norms and economic factors mean that long hours remain an issue in some sectors and workplaces [12, 13].

Within this broader national context, however, working conditions at NTT, and in the research group where I interned, are considerably more flexible. Team members generally enjoy variable daily schedules and the freedom to adjust start and finish times according to experimental needs. Short absences or early departures are handled informally within the team, although the company does maintain an official attendance record to mark the arrival and departure times of each employee. Annual leave entitlements are allocated according to company policy and personal circumstances, and culturally common holidays such as Obon (mid-August) are widely respected [11]. Researchers typically have a good degree of freedom in organizing their workload, provided that they coordinate equipment usage and meet collaborative responsibilities.

NTT has publicly promoted a range of work-style reforms intended to modernize working practices. Remote-work options, flextime systems, and policies designed to encourage employees to actually take paid leave, sometimes described internally as *leisure-style reform*. These initiatives are part of a broader corporate effort to support employees’ physical and mental well-being while improving organizational productivity and agility [6].

As an intern I found that day-to-day flexibility is high: leave and time-off are largely at the discretion of my supervisor, subject to project needs and HR rules.

2.2.3 Diversity, equity and career growth

Japanese corporate promotion systems often differ from those in other countries, combining seniority, performance, and internal career paths. In my group most members are relatively young (under 50), which suggests opportunities for advancement over time. However, promotion timelines and criteria are typically managed through formal internal processes. Gender balance remains a challenge across many parts of Japanese industry. While NTT has public programs to improve diversity and inclusion, most groups still show clear male predominance, with women only occupying secretary or HR roles [9, 10]. As an illustration, all the employees working in my group are only men¹.

2.3 Safety and health

2.3.1 Generalities

Japan places strong emphasis on workplace safety and public hygiene. These priorities protect employee well-being and support high standards of productivity. Workplace safety and health are regulated principally by the Industrial Safety and Health Act, which requires employers to maintain safe and comfortable workplaces through risk assessment, safety management systems, provision of personal protective equipment, mandatory training, and periodic medical checks. In addition, employers must enroll employees in workers’ accident compensation insurance, known as *rōsai hoken*, that covers work- and commute-related injuries and illnesses [14, 15].

Public cleanliness and orderly behavior in cities and corporate campuses further complement occupational safety. For instance, common areas and pedestrian routes are well maintained, and cultural practices, such as wearing masks when ill, can reduce infection risk and demonstrate mutual respect in shared spaces. The NTT Group codifies these expectations in its corporate ethics and safety policies and operates specific systems for disaster preparedness and emergency response (earthquake, heavy rain, pandemic).

To implement these legal and corporate requirements, NTT maintains internal safety and health rules and organizes workplace health and safety committees at sites with sufficient staff. Workplaces with 50 or more regular employees typically have a health and safety committee and access to an industrial physician. These bodies review

¹See the Nanodevices Research Group (Butsude) official website : <https://www.brl.ntt.co.jp/group/sendeg/>

workplace risks, supervise mandatory health checks, and propose measures to prevent occupational accidents and excessive overtime. NTT also operates a Health Management Promotion Committee that coordinates health-promotion measures across Group companies together with the NTT Health Insurance Union. These group-level efforts use PDCA (Plan–Do–Check–Act) cycles to set targets, implement actions and verify outcomes. [10]

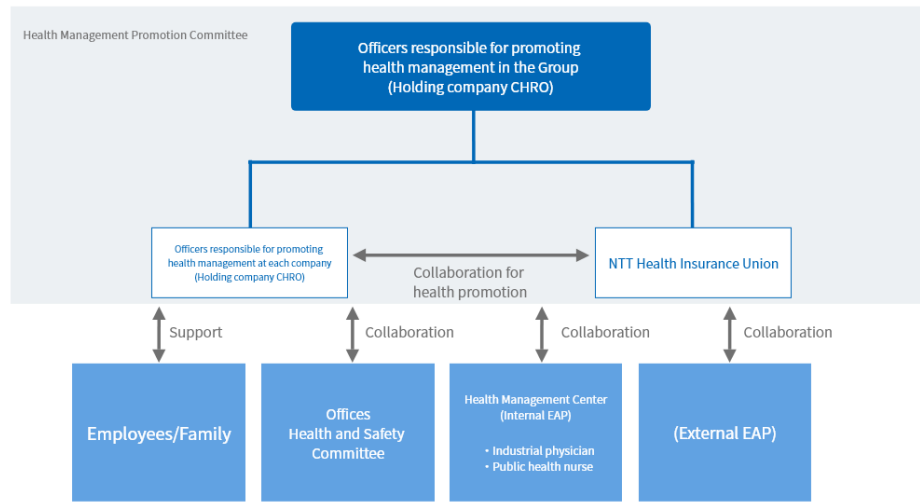


Figure 2.1 – NTT’s Health Management Promotion Committee structure [10].

NTT publishes and practices concrete safety measures: disaster plans, emergency-response headquarters, evacuation drills and the distribution of safety information (in Japanese with English translation where required). The Group’s pandemic response included temporary restrictions on international travel for staff and other measures to reduce infection risk, steps that are documented in company reports from the COVID-19 period.

NTT has been externally recognized for its health-promotion efforts. The Group was selected under METI’s Certified KENKO Investment for Health program, the so-called *White 500* list of outstanding organizations, and has received high-level certification in municipal/state health-promotion recognition schemes, reflecting sustained corporate investment in employee health.



Figure 2.2 – External evaluation agencies of NTT [16, 17].

2.3.2 Safety in daily life and company housing

In some cases, corporate housing or company-arranged accommodation like mine is equipped with professional security services supplied by specialist vendors (SECOM for example), and companies commonly provide information and support to interns and overseas staff about local safety and emergency procedures. As an illustration, if anything wrong is detected in my apartment, the SECOM network will immediately reach out to my supervisor and Otani-san, which is truly comforting in my viewpoint. These private security arrangements and the company’s disaster-preparedness programs are intended to give employees and interns an additional layer of reassurance outside the workplace [10, 18].

2.4 Labor unions and employee representative committees

The NTT Group has labor-union representation organized under the *All NTT Workers Union* (NTT Union), which represents employees across the Group and provides channels for collective bargaining, legal consultation and member services. Union activity and awareness vary between workplaces; large companies commonly have an active union presence, but the visibility of union activity at a specific research site depends on local practice and membership [19].



Figure 2.3 – NTT Union’s brand symbol [19].

Employee representative bodies and workplace committees complement unions by addressing site-level issues such as safety, facilities and staff welfare. The formal responsibilities of these committees and the practical impact they exert on daily working conditions can differ by site. At Atsugi R&D Center, shared facilities and welfare infrastructure such as convenience stores, fitness rooms and sports fields, are visible signs of investment in employee life, even if team-level social events tend to be modest and occasional. In my group, social events come down to welcome/farewell parties and the lunchtime, which in my opinion did not instill in me a strong sense of belonging to the group.

2.5 Corporate governance

NTT maintains a governance framework that separates decision-making and supervisory functions from business execution. The company is structured as a company with an Audit & Supervisory Board and has established voluntary Nomination and Compensation Committees that include outside, independent directors to enhance objectivity and transparency in appointments and pay matters. These governance features sit alongside NTT’s broader risk-management and compliance systems, which incorporate safety and health policies into the Group’s corporate-ethics framework [10].

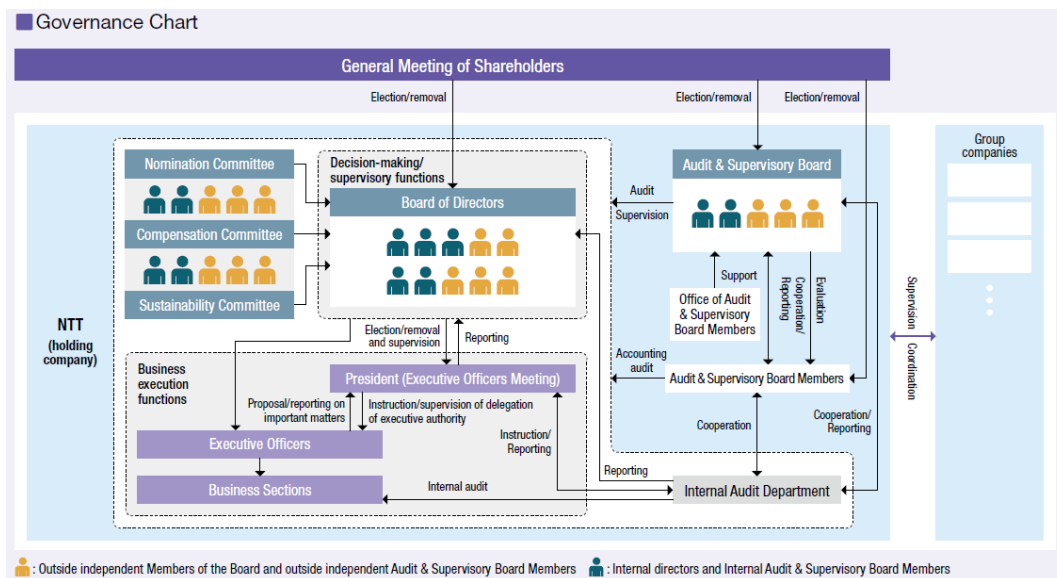


Figure 2.4 – NTT Governance Chart [10].

Chapter 3

Sustainability and environmental considerations

3.1 Sustainability strategy

Sustainability is placed at the core of NTT's corporate strategy and is explicitly linked to the Group's medium-term management plans. In April 2024 the NTT Group revised its Global Sustainability Charter to strengthen alignment with the Medium-Term Management Strategy and to respond to evolving social and environmental expectations. The Charter organizes action around three strategic themes:

1. ensuring the positive coexistence of nature and humanity,
2. improving prosperity for all people and culture,
3. maximizing well-being for all.

These themes are operationalized through four priority categories: climate change, human capital, new value creation and resilience [4, 20].

NTT's R&D organizations, including the IOWN Integrated Innovation Center, the Service Innovation Laboratory Group, the Information Network Laboratory Group, and the Science and Core Technology Laboratory Group, coordinate environmental and energy-related research under a common policy framework. Their activities range from developing low-power communications infrastructure to exploring next-generation materials, with the aim of simultaneously reducing societal CO₂ emissions and enabling new, sustainable services [5, 21].

NTT has set specific operational targets and action items that include improving power efficiency for data transmission, applying energy-conserving designs for data centers and network infrastructure, promoting resource recycling across product life cycles, and supporting ecosystem conservation activities (afforestation, greening, biodiversity initiatives). For example, the Group's Green Bond Framework and related disclosures describe targets for improving power efficiency and making capital investments that reduce environmental footprint across telecommunications and data-center operations [10, 22].

Japan's national strategy of *Society 5.0*, which is a human-centered, cyber-physical societal vision that leverages IoT, AI, robotics and big-data to address social challenges, is a conceptual backdrop for NTT's innovation strategy. NTT positions IOWN and other initiatives as infrastructure elements that can accelerate the emergence of *Society 5.0* and feed into the national innovation ecosystem [23, 24].

3.1.1 Sustainability programs and education

The NTT Group has run an internal Sustainability Conference series since 2013 (expanded as the Group Sustainability Conference) to surface and scale sustainability initiatives across Group companies. NTT also organizes environmental education and capacity-building for staff and environmental coordinators across the Group, offering regular study sessions and training in topics such as SDGs, ESG, energy conservation, waste management and biodiversity stewardship [10].

3.1.2 NTT at Expo 2025

At Expo 2025 Osaka, Kansai taken place from April 13 to October 13, 2025, the NTT Group is a major Pavilion Partner and presented its sustainability vision and technical road map, leveraging IOWN, advanced R&D and ICT demonstrations to illustrate pathways toward a low-carbon, resilient society. The Expo activities are intended to communicate how technologies and platforms can be combined to address Net-Zero and social challenges while engaging the public and international partners [25].



Figure 3.1 – The NTT Group is a Pavilion Partner of Expo 2025 Osaka, Kansai [25].

3.2 Environmental considerations

3.2.1 Concrete actions

NTT applies environmental assessment tools and design standards when developing infrastructure. When proposing new data centers and buildings, the Group uses the Comprehensive Assessment System for Built Environment Efficiency (CASBEE) and aims for the highest CASBEE ranking (S) in order to limit ecological impacts and raise building performance for energy, comfort and sustainability [10, 26].

The company has also formalized governance for environmental management: a Green Innovation Committee chaired by executive leadership, stakeholder engagement processes, and mechanisms to ensure compliance with environmental laws and regulations. At the operational level the Group promotes measures for greenhouse-gas reduction, resource recycling, ecosystem conservation, and energy-conservation R&D as part of its Environmental Target 2030 road map and related disclosures [20, 22].

Within R&D Center campuses and facilities, visible sustainability efforts include recycling and waste-separation stations, extensive campus greening, and the presence of on-site educational materials. That said, large-scale industrial and logistics activities, plus national consumption patterns such as the widespread single-use plastic in food packaging, mean that corporate efforts must be balanced against broader societal practices. Achieving full neutrality remains a long-term challenge that requires supply-chain and societal shifts as well as internal action [10, 20].

Target	Commitment
Scope 1 + 2 reduction (FY2030 vs FY2018)	80% reduction [27].
Scope 1 + 2 + 3 reduction (FY2030 vs FY2018)	40% reduction [27].
Net-Zero target	Net-Zero by FY2040 (SBTi-certified) [4, 27].
Societal contribution target	Enable CO ₂ reductions in society at least 10× greater than NTT’s own emissions (ambition) [10].

Figure 3.2 – Key NTT Group greenhouse-gas and societal contribution targets, validated by the Science Based Targets initiative (SBTi), reflecting alignment with global climate trajectories.

For example, NTT is expanding solar, wind, and distributed energy systems through NTT Anode Energy, while simultaneously upgrading data centers and network equipment to minimize energy consumption. R&D plays a direct

role in these objectives, particularly through research on ultra-low-power photonic networks, efficient semiconductor devices and materials, next-generation cooling technologies, and optimization algorithms for large-scale ICT systems.

As a matter of fact, from September 17 to September 19, 2025, NTT Group companies have participated in the 7th Decarbonization Expo, Japan’s largest ever showcasing corporate initiatives for achieving carbon neutrality. Were presented various solutions that support organizations in advancing their decarbonization goals including NTT Data’s C-Turtle, a greenhouse gas emissions visualization platform.

3.2.2 Personal viewpoint

During my internship, I was not directly confronted with environmental issues, as my work did not explicitly address such concerns. Nevertheless, one observation that can be drawn from my experience is that the dilution refrigerator setup used in my experiments requires a substantial amount of energy, as it is operated continuously for experiments lasting several months, and in some cases several years. Likewise, it is standard practice to keep computers running for extended periods in order to ensure the continuity of these experiments. In this context, the research field in which I was involved naturally raises questions regarding the balance between experimental performance and energy efficiency.

More generally, at the level of Japanese society, waste sorting practices in public spaces, particularly in subway and train stations, may occasionally lack uniformity. Furthermore, the widespread use of plastic and, more broadly, extensive packaging for products that do not always strictly require it suggests the importance of seeking an appropriate balance between hygiene, convenience, and resource efficiency. Finally, the frequent use of air conditioning and heating in public indoor spaces, as well as in public transportation, can at times appear greater than strictly necessary.

Chapter 4

Personal review

4.1 Professional & Scientific experience

This six-month internship at NTT Basic Research Laboratories has been an exceptional and unforgettable experience. It allowed me to discover a professional and social culture very different from that of France, one that places great emphasis on mutual respect and rigor, which are values that create a highly pleasant working environment. Although working conditions in Japan are often described as demanding, NTT proved to be a positive exception. I learned tremendously, both scientifically and personally, through numerous discussions with my supervisor and several colleagues from the Nanodevices Research Group¹.

The facilities and laboratories are well-equipped, ensuring that employees can work in comfortable conditions, even if some spaces occasionally appear slightly cluttered due to intense activity. Scientifically, I had the opportunity to explore a cutting-edge research topic at the interface of quantum physics and electronics. Over the course of six months, I combined extensive literature review with hands-on experimental work, pursuing several complementary directions. My supervisor consistently took the time to explain concepts, provide guidance, and help me make well-reasoned research decisions.

Beyond the laboratory, I was introduced to the daily routine of Japanese office life. At NTT, however, employees are highly skilled professionals operating at the boundary between researcher and engineer. I am fully aware that the level of comfort, safety, and support I enjoyed during these six months does not necessarily reflect the average Japanese workplace. NTT offers an outstanding environment with excellent infrastructure, strict safety and hygiene standards, and well-established financial and administrative support.

4.2 Personal skills

This experience significantly strengthened both my technical and interpersonal competencies. Key skills developed include:

- *Experimental skills*: cryostat operation, signal acquisition, noise reduction, device wiring, low-temperature calibration, and measurement automation.
- *Analytical skills*: Python- and Pyzo-based data processing, numerical fitting, and interpretations.
- *Scientific communication*: preparing clear figures, reporting findings, and articulating hypotheses during discussions with members of the group.
- *Soft skills*: intercultural communication, collaborative problem solving, and time management within a highly autonomous research environment.

These skills will be directly valuable for my future graduate studies, especially in fields involving quantum engineering.

¹See the Nanodevices Research Group (Butsude) official website : <https://www.brl.ntt.co.jp/group/sendeg/>

4.3 Work-life balance

Living conditions were equally remarkable. NTT provided me with fully equipped, rent-free housing in a residence that brought together most international interns (from Canada, the United States, the United Kingdom and European countries). This made it easy to build friendships and share experiences. The residence was also conveniently located within walking distance of a main bus stop, with direct access to Atsugi R&D Center in about 30–40 minutes. On the administrative side, Ms. Mizuki Otani facilitated every aspect of my installation, including opening a bank account, handling utilities (water, electricity, and gas), and managing the SECOM security system. My arrival in Japan was further supported by my supervisor, Dr. Gento Yahamata, who kindly helped me discover the city and settle into my new environment.

My supervisor was also very flexible in granting vacation days, which gave me the chance to travel across Japan without concern (Tokyo, Mount Fuji, Mount Oyama, Hakone, Yokohama, Kamakura, Enoshima, Hiratsuka, Nabari, Nikko, Osaka, Nara, Kyoto, Nagoya, Nagano and more) and also in other Asian countries such as China (Hong Kong, Guangzhou) and Vietnam (Hanoi, Ninh Binh, Cat Ba, Ha Long Bay). Public transportation in Japan is extremely reliable and comfortable, and my interactions with Japanese people were consistently respectful and pleasant. Despite my limited command of the language, many locals took the time to engage with me and answer my questions. I found the Japanese people to be remarkably welcoming, devoted to their work, and deeply respectful of one another. Combined with Japan's rich culinary traditions and strong cultural heritage, these aspects made my stay truly unforgettable.

4.4 Future prospects

From a scientific perspective, research on quantum physics applied to fundamental electronics turned out to be surprisingly enjoyable, even though I was not originally inclined toward electronics or signal processing. Applying quantum theory in experimental contexts was both challenging and rewarding, and it has strengthened my desire to deepen my knowledge in both theoretical and experimental quantum physics. Looking ahead, I hope to build on these skills to pursue further research in quantum chemistry, an emerging field at the edge between molecular chemistry and quantum dynamics. As a result, I have already applied for PhD programs in the United States in Quantum Science and Engineering at Harvard and Princeton, and in Physical Chemistry at MIT, and I plan to apply for MSc at EPFL, ENS and Ecole Polytechnique in quantum engineering.

The technical and conceptual foundations built at NTT will support my applications and prepare me for research environments requiring precision, creativity, and deep physical insight. This experience marks a decisive step in my academic development and long-term career aspirations.

This was my very first six-month corporate internship, and it will undoubtedly remain one of the most rewarding professional experiences of my career. I am sincerely grateful to NTT BRL and ESPCI Paris-PSL for giving me this invaluable opportunity.

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